United States Patent Application Publication 20250270910
Kind Code A1
Publication Date August 28, 2025
Inventor(s) Cote; Brennon Leigh et al.

Regenerative Pumping System And Method

Abstract

A system is provided for determining piston position and for monitoring and maintaining stroke rate in a hydraulic lift system comprising a lift cylinder, a reversible hydraulic pump, a motor and a flywheel. A method is further provided for managing energy requirements of a hydraulic lift system comprising a lift cylinder, a reversible hydraulic pump, a motor and a flywheel.

Inventors: **Cote; Brennon Leigh (Cochrane, CA), Twidale; Derek Lorne (Cochrane, CA), Dyck; David Paul (Cochrane, CA)**

Applicant: **ARTIFICIAL LIFT PRODUCTION INTERNATIONAL CORP. O/A AL PRO INTERNATIONAL CORP.** (Cochrane, CA)

Family ID: **96659577**

Appl. No.: **18/585875**

Filed: **February 23, 2024**

Publication Classification

Int. Cl.: **E21B43/12** (20060101); **F04B47/04** (20060101); **F04B49/24** (20060101)

U.S. Cl.:

CPC **E21B43/129** (20130101); **F04B47/04** (20130101); **F04B49/24** (20130101);

Background/Summary

FIELD

[0001] The present disclosure relates to a improvements to regenerative pumping systems in connection with artificial lift.

BACKGROUND

[0002] In oil production, oil and liquid hydrocarbons are often raised to surface using systems called artificial lift systems. The systems typically use a downhole pump to pump liquid hydrocarbon from underground formations in a cased wellbore up to surface. A piston on the downhole pump is connected via a rod string to an actuator at the surface to thereby reciprocate the piston of the downhole pump to raise fluid to surface. The actuation used to create the reciprocating motion can be done in several ways, the most common being the classic mechanical pumpjack.

[0003] The reciprocating motion can also be achieved by a hydraulic pumpjack, comprised of a lift cylinder and a power unit. A piston of the hydraulic lift cylinder is threadably connected to the rod string. Hydraulic fluid and pressure applied to one side of the piston serves to raise the lift cylinder piston, the rod string, and the piston of the downhole pump, in turn lifting the fluids, on the upstroke. On the downstroke, the cylinder piston lowers with weight of the rod string and the downhole pump piston is lowered within the downhole pump.

[0004] Piston stroke speed and piston position, which directly relate to downhole pumping speed, is a very important aspect and measure of the efficiency and rate of production of the well. Therefore tracking piston stroke speed and position can assist with ensuring that piston stroke speed is meeting requirements for piston strokes per minute and daily production quotas.

[0005] A motor powers a hydraulic pump capable of delivering the hydraulic fluid from a hydraulic reservoir to lift the lift cylinder in the upstroke. Gravity and downward motion of the rod string channels fluid from the lift cylinder in the downstroke, requiring no motor power on the downstroke.

[0006] Various techniques have been examined for collecting power from the downstroke for adding to motor power on the upstroke:

[0007] U.S. Pat. No. 5,647,208A teaches a flywheel-motor-hydraulic pump arrangement in which a swash plate of the hydraulic pump controls the operation of the hydraulic pump in pump (upstroke) or motor (downstroke) mode. The swash plate and corresponding control system also control the speed of the hydraulic pump in both modes. U.S. Pat. No. 7,234,298B2 teaches a dual hydraulic pump/loop system in which one hydraulic pump/loop is used on the upstroke and another is connected to the flywheel on the downstroke. U.S. Pat. No. 5,827,051 teaches a regenerative hydraulic power transmission system including means to reverse flow of hydraulic fluid through the hydraulic pump and an inertial assist for the power source that gathers energy in the downstroke and utilizes the gathered energy to power the upstroke. WO2013/093511 teaches a hydraulic system including a flywheel connected to a hydraulic pump for powering a hydraulic actuator.

[0008] However, most systems experience significant loss of efficiencies, as well as speed limitations for both the hydraulic pump and the motor. As well many systems require complex energy capture systems, which are expensive and hard to manage, as well as present further losses in energy capture efficiency.

[0009] A need further exists for more precise tracking and control of regenerative pump systems to ensure flywheel speed is being optimized, and that piston stroke rate is meeting production requirements.

SUMMARY

[0010] A system is provided for determining piston position and for monitoring and maintaining stroke rate in a hydraulic lift system comprising a lift cylinder, a reversible hydraulic pump, a motor and a flywheel. The system comprises an ultrasonic sensor located on an upstroke side of a piston of the lift cylinder for providing piston position data; and a computer implemented program for conducting the steps of: [0011] i. estimating a position of the piston from swash plate position data of a swash plate of the hydraulic pump and from speed of the motor; [0012] ii. estimating, from pump pressure data, a flow rate of fluid lost to the system through internal leakage in the pump; [0013] iii. correcting, using the leakage-compensated estimated fluid flow rate and from dimensions of the lift cylinder, any inaccuracies in the estimated position of the piston; and [0014]

iv. receiving position data from the ultrasonic sensor and comparing with the corrected estimated position of the piston.
[0015] If the ultrasonic sensor piston data is determined to be erroneous, or if no reading is available from the ultrasonic sensor, then the estimated piston position is adopted as the correct piston position.
[0016] A method is further provided for managing energy requirements of a hydraulic lift system comprising a lift cylinder, a reversible hydraulic pump, a motor and a flywheel.
[0017] The method comprises the steps of: [0018] a. detecting a new stroke of a piston of the lift cylinder; [0019] b. determining an optimal flywheel speed for the stroke, optimal flywheel speed being a minimum speed setpoint at which capacity of the hydraulic pump is not exceeded at any point in the stroke; [0020] c. applying a design margin to the optimal flywheel speed; [0021] d. determining: [0022] i. if the motor has sufficient power to reach the optimum flywheel speed during the piston stroke, then [0023] ii. setting parameters of new piston stroke as defined by user's inputs; or [0024] iii. if the motor power is insufficient to reach the optimum flywheel speed, then [0025] iv. setting a target flywheel speed to the optimum flywheel speed, and setting parameters of the new piston stoke to match parameters from a previous stroke; and [0026] e. repeating steps a. through d. until the motor power is sufficient to reach the flywheel optimum speed.
[0027] A system is further provided for managing energy requirements of a hydraulic lift system comprising a lift cylinder, a reversible hydraulic pump, a motor and a flywheel. The: system comprises a computer implemented program for conducting the steps of the method of described above.
[0028] A back pumping piston for use in a piston-cylinder, comprising one or more oil intake holes formed in a body of the back pumping piston; a central bore within the back pumping piston in fluid communication with the one or more oil intake holes, said central bore including a high-pressure seal; a high-pressure cavity in fluid communication with the central bore and comprising a check valve to selectively prevent outflow of oil from the high-pressure cavity; a low-pressure cavity defined at a lower end by a plunger and being in fluid communication with a gas chamber above the back pumping piston; and a return spring connected to the plunger to bias the plunger into the low-pressure cavity; Receipt of compressed gas from the gas chamber above the piston into the low-pressure cavity serves to overcome the bias force of the return spring and to lower the plunger through the central bore until a lower end of the plunger engages with the high-pressure seal, thus sealing off the oil intake holes and preventing exit of oils through the oil holes and compressing oils within the high-pressure cavity until oil pressure within the high-pressure cavity is sufficient to overcome the check valve, allowing oils to pass through the check valve and out into a hydraulic oil circuit.
[0029] A rod seal assembly is also provided, for use with a rod connected to a piston-cylinder of an artificial lift system, said rod having multiple rod seals and an oil scraper. The rod seal assembly comprises: [0030] a. a gas inlet connection connectable to a pneumatic source for receiving compressed gas; [0031] b. a gas inlet passage in communication with the gas inlet connection for receiving compressed gas and passing said compressed gas through oil holes formed in the rod seal gland of the rod seal assembly, to thus forcing out any oil within the oil holes; and [0032] c. an oil outlet passage for receiving oil forced out of the oil holes and passing it through an oil outlet connection connectable to the gas volume above the piston in the cylinder.
[0033] The oil outlet passage is sized such that, once the oil outlet passage is filled with oil, any gas pushing the oil is prevented from bypassing the oil to the oil outlet connection.
[0034] A hydraulic lift system is further provided, comprising a lift cylinder, a reversible hydraulic pump, a motor and a flywheel, wherein said system comprises downstroke bypass valve in fluid communication with a lower chamber of the lift cylinder, said downstroke bypass valve being openable to allow hydraulic fluid from the lower chamber of the lift cylinder to bypass the hydraulic pump and flow directly to a reservoir.

[0035] It is to be understood that other aspects of the present disclosure will become readily apparent to those skilled in the art from the following detailed description, wherein various embodiments of the disclosure are shown and described by way of illustration. As will be realized, the disclosure is capable of other and different embodiments and its several details are capable of modification in various other respects, all without departing from the spirit and scope of the present disclosure. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not as restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0036] A further, detailed, description of the disclosure, briefly described above, will follow by reference to the following drawings of specific embodiments of the disclosure. The drawings depict only typical embodiments of the disclosure and are therefore not to be considered limiting of its scope. In the drawings:
[0037] FIG. **1** is a perspective view of one embodiment of a system of the present disclosure;
[0038] FIG. **2**A is a schematic diagram of the system of FIG. **1**, showing fluid flow in a cylinder upstroke;
[0039] FIG. **2**B is a schematic diagram of the system of FIG. **1**, showing fluid flow in a cylinder downstroke;
[0040] FIG. **3**A is a schematic diagram of a control and feedback system of the present disclosure;
[0041] FIG. **3**B is a schematic diagram of the energy balance and flywheel speed optimization methods of the present system;
[0042] FIG. **4** logic flow diagram of a feedback and correction system in relation to measuring piston position in the present disclosure;
[0043] FIG. **5**A is a top end view of a back pumping piston of the present disclosure;
[0044] FIG. **5**B is a cross-sectional elevation view of the back pumping piston of FIG. **5**A, in an uncompressed state;
[0045] FIG. **5**C is a cross-sectional elevation view of the back pumping piston of FIG. **5**A, in a compressed state;
[0046] FIG. **6**A is a top plan view of a rod seal assembly of a rod seal oil return system of the present disclosure;
[0047] FIG. **6**B is a cross-sectional elevation view of the rod seal assembly of FIG. **6**A;
[0048] FIG. **7**A is a front elevation view of a pumping assembly of the rod seal oil return system;
[0049] FIG. **7**B is a side cross-sectional elevation view of the pumping assembly of FIG. **7**A; and
[0050] FIG. **7**C is circuit diagram of the rod seal oil return system.
[0051] The drawings are not necessarily to scale and in some instances, proportions may have been exaggerated in order to more clearly depict certain features.
DETAILED DESCRIPTION
[0052] The description that follows and the embodiments described therein are provided by way of illustration of an example, or examples, of particular embodiments of the principles of various aspects of the present disclosure. These examples are provided for the purposes of explanation, and not of limitation, of those principles and of the disclosure in its various aspects.
[0053] With reference to FIGS. **1** and **2**, the present disclosure relates to improvements in the control and efficiency of a hydraulic pumpjack which includes a regenerative system **100** working in relation to a hydraulic lift cylinder **18**. An example of such a regenerative system **100** includes a motor or power source **10**, a hydraulic pump **14** capable of delivering hydraulic fluid from a reservoir **6**, through the isolation valve **15**, to the hydraulic lift cylinder **18**. The hydraulic pump **14** is reversible so as to channel fluid from the lift cylinder **18** in a cylinder downstroke, back to the

reservoir **6**.

[0054] The hydraulic pump **14** is preferably of an open-loop, over-center, electronic displacement control design which allows the hydraulic fluid to flow in both directions through the hydraulic pump **14** to and from the lift cylinder **18** and reservoir **6**, creating the reciprocating motion of the hydraulic lift cylinder **18**.

[0055] The present system **100** further includes a flywheel counterbalance **12** to collect energy on the downstroke. The flywheel counterbalance **12** is connected to the motor shaft **8** and the motor shaft **8** is connected to power the hydraulic pump **14** via a step-down/step-up system **16** such as a pulley sheave or a belt drive system. The arrangement allows the motor **10** to have only a single output shaft **8**. A swash plate within the hydraulic pump **14** creates the reversibility, converting the hydraulic pump into a hydraulic motor on the cylinder downstroke, to rotate the flywheel **12** and transfer energy recovered from the down stroke, into the flywheel **12**.

[0056] The motor **10** and flywheel **12** are preferably able to operate at a common, high speed. The hydraulic pump **14** can operate at a lower speed-the step-down/step-up system **16** acts as a transmission to step up or step down the speed depending on which direction fluid is travelling.

[0057] In the present system **100**, the motor **10** is directly connected to the flywheel **12** via motor shaft **8** while the hydraulic pump **14** is connected to the flywheel **12** and via step-down/step-up system **16**. This allows the flywheel **12** and the hydraulic pump **14** to operate at different speeds. In the present open loop over-center pump circuit, as illustrated in FIGS. **2**A and **2**B, the hydraulic fluid travels from the reservoir **6**, through the hydraulic pump **14**, to the cylinder **18**, and then is returned via the same set of piping to the reservoir **6** via the hydraulic pump **14**, there is no requirement for separate sets of lines for sending fluid from reservoir to the lift cylinder vs sending fluid back to the reservoir from the lift cylinder.

[0058] In operation, on a first upstroke of the hydraulic lift cylinder **18**, the motor **10** provides initial power to the system **100** to get the flywheel operating at an optimal speed. The flywheel, once operating at optimal speed, then powers the hydraulic pump **14** to pump fluid from reservoir **8** to lift a piston of hydraulic cylinder **18**. The amount of energy that the motor **10** is required to supply to the system **100** for a particular stroke is calculated such that, at a bottom of a cylinder stroke, the flywheel **12** has reached a target speed. Energy may be input from the motor **10** into the system at any point in the stroke and the torque or power may vary during the stroke. Typically, the motor **10** inputs this energy as a constant torque or power throughout the entire stroke. The constant torque application equates to a constant current draw on the motor **10**, ultimately minimizing power costs and peak current draw. As the flywheel **12** expends energy to lift the rod string, the energy required to lift the lift cylinder **18** is greater than the energy supplied by the motor **10** and the flywheel decreases in speed to a minimum speed at the top of stroke.

[0059] On the downstroke of the hydraulic cylinder **18**, the swash plate in the hydraulic pump **14** reverses flow of fluid through the pump **14** and converts the pump **14** to hydraulic motor operation mode. The potential energy of the suspended rod string is used to pull the piston of the lift cylinder **18** down, driving the hydraulic pump **14**, now in motor operation mode, taking advantage of the step up/step down system **16**, to accelerate the flywheel **12** via shaft **8**. Motor **10** continues to apply torque to the flywheel **12** also causing the flywheel **12** to accelerate. The combined application of energy by the motor **10** and the hydraulic pump **14** in hydraulic motor operation mode result in the flywheel **12** reaching its optimal speed.

[0060] The present hydraulic lift cylinder **18** is preferably a single acting, piston down design where the piston of the hydraulic cylinder **18** is directly connected to the rod string of the downhole pumping system. It should be noted that the current regenerative system **100** can be used with other hydraulic lift cylinder designs.

Control System

[0061] With References to FIG. **3**A an overall schematic diagram is provided for a method of controlling a hydraulic pumpjack with hydraulic lift cylinder **18**. All of the steps performed in the methods of FIGS. **3**A and **3**B are programmed into either a local or remote programmable logic computer (PLC).

[0062] At startup the stroke position, stroke length and stroke rate are input into the control system from various sensors and by the operator via a human-machine-interface (HMI) that can be a computer, touchscreen or other suitable interface. A new stroke is detected when the cylinder **18** reaches its bottom of stroke position, when the stroke time has elapsed or when the control system is initially started. When a new stroke is detected then an Energy Balance is performed to ensure sufficient energy is being input by the motor **10** to meet the energy requirements of a next subsequent stroke. The Energy Balance methodology is illustrated in more detail in FIG. **3**B and discussed in further detail below.

Energy Balance and Flywheel Speed Optimization

[0063] When a new stroke is detected, the present systems and methods perform an analysis to determine an optimal flywheel speed for the upcoming stroke. The optimal flywheel speed is defined as the minimum speed setpoint where the pump capacity will not be exceeded at any point in the stroke. A design margin may then be applied to the optimal flywheel speed. If the motor has sufficient power to reach the optimum speed during the stroke, the next stroke will occur as defined by the user's inputs, if the power is not sufficient then the target flywheel speed will be set to the optimum flywheel speed, as calculated above, and the stoke parameters from the previous stroke will be used. This method is repeated until the motor power is sufficient to reach the flywheel target speed. As a first step in the process, the energy consumed on the previous stroke and the hydraulic efficiency is calculated, as illustrated in FIG. **3**B where: [0064] E=energy [0065] FW=flywheel [0066] SOS=start of stroke [0067] EOS=end of stroke [0068] DN=down [0069] Hyd_eff=hydraulic efficiency

[0070] From that calculation, along with any new user inputs from the HMI regarding the cylinder stroke speed, stroke length, dwell times and stroke symmetry, the minimum flywheel speed at the top of the stroke can be calculated. Next, the minimum flywheel speed at the bottom of the stroke can be determined. A check is next performed in the method to determine if the motor can put enough energy into the flywheel on the next stroke to obtain the minimum flywheel speed at the bottom of the stroke. If this is confirmed then the new user inputs are immediately implemented. If the motor cannot put enough energy into the flywheel on the next stroke, the parameters from the last stroke are maintained with the new flywheel minimum speed setpoint until it is determined that the motor is able to put enough energy into the flywheel.

[0071] The lift cylinder position is then determined from the Position Prediction/Correction algorithm FIG. **4**. This will be discussed in further detail below.

Position Read and Position Prediction and Correction

[0072] With reference to FIG. **4**, the present disclosure further provides a prediction and correction system for making an accurate determination of the position of the lift cylinder piston as it is reciprocating within the hydraulic cylinder. Piston position determination is used by the Control System of the present system, as feedback to determine if the cylinder is correctly following the target piston position generated by the calculated cylinder piston path.

[0073] There are several known methods of measuring the position of a piston of a hydraulic cylinder while it is in motion. These generally involve using some type of linear position sensor such as a hall effects type sensor with a sensing rod and a magnet on the moving portion to measure position. These types of sensor work well over short cylinder stroke lengths, but over longer stroke lengths they become less reliable, impractical to install or replace, and cost prohibitive. Another method is to use a flow meter to monitor the amount of hydraulic fluid pumped into the cylinder and make an estimate of the position using the dimensions of the cylinder. The issue with this method is that inaccuracy of the flow meters lead to a constant drift between the actual position of the piston and the estimated position. More accurate flow meters could be utilized but they become cost prohibitive as well. Another problem with all of these methods is that any erroneous reading from the position sensor or flow meter can result in erratic behavior from the cylinder control system and any momentary loss of signal will cause a system fault and result in a shut-down.

[0074] The present disclosure provides piston position sensing in the form of a non-contact ultrasonic sensor **60**. The ultrasonic distance sensor is installed in the end of the cylinder to sense the position of the piston inside the cylinder. These ultrasonic sensors are compact, accurate and cost effective. One issue with ultrasonic sensors is that they can become less accurate at high pressures, and cease to work at very low pressures, such as those pressures experienced with the lift cylinder at the top of an upstroke or bottom of a downstroke respectively.

[0075] Simultaneous to position sensing by the ultrasonic sensor, the present prediction and correction system calculates an estimated position of the piston using swash plate position data from a position sensor on the hydraulic pump **14** swash plate, and data from the speed of the motor **10** to predict the nominal amount of fluid passing through the pump **14**. This estimated position of the lift cylinder piston is made more accurate by using pump pressure sensors to accurately estimate the flow rate of fluid lost to the system through internal leakage in the pump as well as the pump and other control valves in the system. This leakage compensated flow rate is used with the dimensions of the cylinder to calculate the estimated position of the piston.

[0076] Software within the present system analyzes the reading from the ultrasonic sensor and compares it with the estimated position as calculated above based on swash plate position and motor speed. Based on the comparison, if the system determines the ultrasonic sensor result to be erroneous, or if for any reason there is no reading from the ultrasonic sensor, then the present system uses the estimated position instead. This eliminates faults and data gaps that can occur from momentary loss of signal from the ultrasonic sensor or erratic behavior of the ultrasonic sensor as increased pressure is experienced in the upstroke or decreased pressure is experienced in the down stroke.

[0077] Although an ultrasonic sensor is used in the preset disclosure, it should be noted that this system would work to verify data from any position sensor.

[0078] The target piston position at any given time is calculated for each iteration of the control cycle based on a targeted path of piston travel, which in turn is calculated from the target stroke rate, stroke length, dwell time at top of stroke and bottom of stroke respectively, and ratio of down stroke time to up stroke time. The accelerations at the top and bottom of each stroke may be calculated in several ways, including sinusoidal position and constant acceleration. A control system, labeled Control in FIG. **3**A, compares the piston position from the Position Read and Error Check method and system and uses a proportional-integral-derivative (PID) controller to send signals to the hydraulic pump swash plate control valves and downstroke bypass valve to move the lift cylinder piston position to follow the target position path.

[0079] In the Setpoint Limits step of the overall control system and method, a calculation is performed for the limit of every input in the system. These are dynamic limits, based on the current operating conditions and any new user inputs, so they must be re-calculated every iteration. These limits are then updated in the PLC.

[0080] The HMI is then updated with updated piston position, speed and efficiency data, as well as physical operating conditions (oil temp, hydraulic pressure, filter status, bearing temps, cooler status, circulating pump status).

Back Pumping Piston

[0081] In the movement of most piston-cylinders, there is often at least some transfer of hydraulic oil that tends to form an oil film between the high-pressure and low-pressure sides of the piston seals. This oil film builds up over time until there is a small volume of oil on top of the piston. In such cases the ultrasonic sensor **60** is not able to tell the difference between the top of the piston itself and the top of this oil volume level and tends to misread the position of the piston by the amount of the oil level on top of the piston. In order to remove the oil volume from on top of the

piston, a back pumping piston **80** has been developed.
[0082] With reference to FIGS. **5**A to **5**C, the back pumping piston **80** works on the principle of applying a low pressure to a large area that is in turn used to create a much higher pressure in another, smaller area. The area ratio that is practical for this back pumping piston **80** is between approximately 10:1 and 400:1 and more preferably about 280:1.
[0083] In operation, on the up stroke of the back pumping piston **80**, the oil film that has passed the piston seals **81** is wiped by the top seal **81**A into the oil intake holes **82** in the piston body. The oil flows from the oil intake holes **82** and the central bore **91** of the plunger **90** into a bore of a central component **84** of the back pumping piston **80**. Oil travels past a high-pressure seal **94** and enters a high-pressure cavity **86**. The high-pressure cavity **86** is open at the oil entry end so that the oil can easily fill it without trapping gas bubbles. A check valve **96** selectively prevents outflow of oils from the high-pressure cavity through a high-pressure exhaust **98** beyond.
[0084] A low-pressure cavity **88** is also present in the back pumping piston **80** and is defined at a lower end by a plunger **90**. The plunger **90** is biased into the low-pressure cavity **88** by a spring **92**, the spring being on one side of the plunger **90** and the low-pressure cavity **88** being on another. As the back pumping piston **80** travels upwards gas above the back pumping piston **80** within the cylinder (not shown) begins to compress and once the gas pressure above the back pumping piston **80** exceeds a pressure in the low-pressure cavity **88**, a pre-load force of the spring **92** is overcome and the plunger **80** begins to move downwards, thereby enlarging a volume of the low-pressure cavity **88**.
[0085] A lower end of the plunger **90** engages with the high-pressure seal **94**, thus sealing off the oil intake holes **82** and preventing their re-exit out of the piston **80** and back out to the rest of the cylinder (not shown).
[0086] The oils inside the high-pressure cavity **86** are then compressed by the lower end of plunger **90**. As the piston **80** continues to rise in the cylinder, and as gas pressure in the low-pressure cavity **88** builds, the plunger **90** continues to lower and the pressure inside of the high pressure cavity **86** also increases until the pressure in the high-pressure cavity **86** becomes greater than a pressure on the other side of the check valve **96** allowing the oils inside the high pressure cavity **88** to pass through the check valve **96**, out of the high pressure exhaust **98** and back into the hydraulic oil circuit (not shown).
[0087] When the piston **80** travels back to the bottom of the stroke, gas pressure in the low-pressure cavity **88** is reduced and the spring **92** biases the plunger **90** to its original position. The return spring **92** is sized so that it can pull the plunger from the high-pressure cavity **86** against full vacuum pressure of the exhaust which enables the piston **80** to operate with a single check valve **96** and not the typical two required for most intensifiers. It would, of course, be well understood by a person of skill in the art, that a dual check valve arrangement would work equally as well with the present system. A regulator provides a means of ensuring that pressure above the piston is low enough so as to not interfere with the ultrasonic sensor **60**, while also ensuring enough pressure for the back pumping piston **80** to operate.
[0088] The back pumping piston **80** allows for oil to be easily pumped from the top of the piston **80** at the top of stroke and back to the hydraulic oil circuit, even when the hydraulic oil circuit is operating at its maximum pressure. The high-pressure cavity **86** is sized to have an un-swept length similar to or greater than the swept length. This is done so that if any gas is present in a significant amount within the high-pressure cavity **86**, then the lower end of the plunger **90** will compress that gas but be unable to force it through the check valve **96** and into the hydraulic oil circuit. The high-pressure cavity **86** is also vertically oriented so that any entrapped gas will preferentially rise and leave the high-pressure cavity **86** and exit through upper oil intake holes **100** formed in the plunger **90**.
Rod Seal Design
[0089] A common issue with the hydraulic cylinder in hydraulic pumping units is that there is a

small amount of hydraulic fluid or oil that bypasses the rod seals in the form of an oil film that is difficult to remove completely with the rod seals. With existing rod seals it is possible to partially back pump this oil film back into the hydraulic oil circuit, using specific seal profiles however they are not 100% effective.

[0090] The present rod seal design uses a pneumatic-over-hydraulic pumping system in combination with a scraper to remove the oil film from the rod and pump it into the un-swept volume of the cylinder where it is pumped back into the hydraulic oil circuit using the back pumping piston **80** configuration described above. The present rod seal design is made up of two separate assemblies shown in FIGS. **6**A/**6**B and in FIGS. **7**A/**7**B respectively.

[0091] With reference to FIGS. **6**A and **6**B, a rod seal assembly **110** is illustrated with a rod **106** of the rod string and multiple rod seals **108** and an oil scraper **122**.

[0092] A gas inlet connection **112** of the rod seal assembly **110** is connectable to the gas volume in the cylinder (not shown) above the piston **80** using a first tube (not shown). An oil outlet connection **114** is also connectable to the gas volume above the piston **80** using a second tube (not shown). When the gas pressure above the piston **80** rises during an upstroke of the piston **80**, gas is forced along the first tube into the gas inlet connection **112**, via a gas inlet passage **118** and through oil holes **116** formed in the rod seal gland **111** of the rod seal assembly **110**. The oil holes **116** are in communication with a cavity **116**A that surrounds the rod **106**. If there is oil inside the oil holes **116** it will be forced through them and into the oil outlet passage **120**. The oil outlet passage **120** is sized such that, given the viscosity of the oil, once the oil outlet passage **120** is filled with oil, the gas pushing the oil will not easily bypass the oil, but instead carries the oil into an inlet **132** of the pumping assembly **130**.

[0093] The pumping assembly **130** is shown in FIGS. **7**A,**7**B and **7**C. The oil passes a first check valve **134** and flows into a single acting cylinder **136** which is caused to extend and in turn to compress a return spring **138** connected to a distal end of the single acting cylinder **136**. When the pressure above the piston **80** decreases on the down stroke the return spring **138** is allowed to extend, and in turn forces the single acting cylinder **136** to compress the fluids inside of it and push them through the second check valve **140** and out an outlet **142** which connects to the oil outlet connection **114** that returns the oil to the cylinder tube above the piston **80**. This cycle continues removing unwanted oil from the cavity **116**A between the oil scraper **122** and the rod seals **108**.

Downstroke Bypass Valve

[0094] During the downstroke of the piston within the cylinder **18**, wellbore conditions may allow the downhole pump to fall faster than the surface hydraulic pump **14** will allow. If this is the case and an increase in pumping speed is required, a downstroke bypass valve **17** can be opened to allow some of the hydraulic fluid returning from a lower chamber of the cylinder **18** to bypass the hydraulic pump **14** and flow directly to the reservoir **6**, thereby allowing for the increasing the velocity of the downstroke.

[0095] The previous description of the disclosed embodiments is provided to enable any person skilled in the art to make or use the present disclosure. Various modifications to those embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments without departing from the spirit or scope of the disclosure. Thus, the present disclosure is not intended to be limited to the embodiments shown herein, but is to be accorded the full scope consistent with the claims, wherein reference to an element in the singular, such as by use of the article “a” or “an” is not intended to mean “one and only one” unless specifically so stated, but rather “one or more”. All structural and functional equivalents to the elements of the various embodiments described throughout the disclosure that are known or later come to be known to those of ordinary skill in the art are intended to be encompassed by the elements of the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims.

Claims

1. A system for determining piston position and for monitoring and maintaining stroke rate in a hydraulic lift system comprising a lift cylinder, a reversible hydraulic pump, a motor and a flywheel, said system comprising: a. an ultrasonic sensor located on an upstroke side of a piston of the lift cylinder for providing piston position data; and b. a computer implemented program for conducting the steps of: i. estimating a position of the piston from swash plate position data of a swash plate of the hydraulic pump and from speed of the motor; ii. estimating, from pump pressure data, a flow rate of fluid lost to the system through internal leakage in the pump; iii. correcting, using the leakage-compensated estimated fluid flow rate and from dimensions of the lift cylinder, any inaccuracies in the estimated position of the piston; and iv. receiving position data from the ultrasonic sensor and comparing with the corrected estimated position of the piston, wherein, if the ultrasonic sensor piston data is determined to be erroneous, or if no reading is available from the ultrasonic sensor, then the estimated piston position is adopted as the correct piston position.
2. The system of claim 1, further comprising a computer implemented control program for conducting the steps of: comparing the correct piston position at a given time to a target piston position along a target position path at that given time; and sending a signal to control valves of the swash plate and to a downstroke bypass valve to move the lift cylinder piston position to follow the target position path.
3. A method for managing energy requirements of a hydraulic lift system comprising a lift cylinder, a reversible hydraulic pump, a motor and a flywheel and the system of claim 1, said method comprising the steps of: a. detecting a new stroke of a piston of the lift cylinder; b. determining an optimal flywheel speed for the stroke, optimal flywheel speed being a minimum speed setpoint at which capacity of the hydraulic pump is not exceeded at any point in the stroke; c. applying a design margin to the optimal flywheel speed; d. determining: i. if the motor has sufficient power to reach the optimum flywheel speed during the piston stroke, then ii. setting parameters of new piston stroke as defined by user's inputs; or iii. if the motor power is insufficient to reach the optimum flywheel speed, then iv. setting a target flywheel speed to the optimum flywheel speed, and setting parameters of the new piston stoke to match parameters from a previous stroke; and e. repeating steps a. through d. until the motor power is sufficient to reach the flywheel optimum speed.
4. The method of claim 3, further comprising the following step prior to step a.: a. calculating energy consumed on a stroke previous to the new stroke the hydraulic efficiency.
5. The method of claim 3, further comprising calculating the minimum flywheel speed at a top of a stroke and at a bottom of a stroke from energy consumed on the previous stroke, hydraulic efficiency, cylinder stroke speed requirements, stroke length, dwell times and stroke symmetry.
6. A computer implemented program for conducting the steps of the method of claim 3.
7. A back pumping piston for use in a piston-cylinder, comprising: a. one or more oil intake holes formed in a body of the back pumping piston; b. a central bore within the back pumping piston in fluid communication with the one or more oil intake holes, said central bore including a high-pressure seal; c. a high-pressure cavity in fluid communication with the central bore and comprising a check valve to selectively prevent outflow of oil from the high-pressure cavity; d. a low-pressure cavity defined at a lower end by a plunger and being in fluid communication with a gas chamber above the back pumping piston; and e. a return spring connected to the plunger to bias the plunger into the low-pressure cavity; wherein receipt of compressed gas from the gas chamber above the piston into the low-pressure cavity serves to overcome the bias force of the return spring and to lower the plunger through the central bore until a lower end of the plunger engages with the high-pressure seal, thus sealing off the oil intake holes and preventing exit of oils through the oil holes and compressing oils within the high-pressure cavity until oil pressure within the high-

pressure cavity is sufficient to overcome the check valve, allowing oils to pass through the check valve and out into a hydraulic oil circuit.

8. The back pumping piston of claim 7, wherein return spring is sized to achieve sufficient force to pull the plunger from the high-pressure cavity against a full vacuum pressure.

9. The back pumping piston of claim 8 wherein the area ratio of plunger top area to lower end area is between 10:1 and 400:1

10. The back pumping piston of claim 9 wherein the area ratio is 280:1.

11. A rod seal assembly for use with a rod connected to a piston-cylinder of an artificial lift system, said rod having multiple rod seals and an oil scraper; said rod seal assembly comprising: a. a gas inlet connection connectable to a pneumatic source for receiving compressed gas; b. a gas inlet passage in communication with the gas inlet connection for receiving compressed gas and passing said compressed gas through oil holes formed in the rod seal gland of the rod seal assembly, to thus forcing out any oil within the oil holes; and c. an oil outlet passage for receiving oil forced out of the oil holes and passing it through an oil outlet connection connectable to the gas volume above the piston in the cylinder, wherein the oil outlet passage is sized such that, once the oil outlet passage is filled with oil, any gas pushing the oil is prevented from bypassing the oil to the oil outlet connection.

12. A hydraulic lift system comprising a lift cylinder, a reversible hydraulic pump, a motor and a flywheel, wherein said system comprises downstroke bypass valve in fluid communication with a lower chamber of the lift cylinder, said downstroke bypass valve being openable to allow hydraulic fluid from the lower chamber of the lift cylinder to bypass the hydraulic pump and flow directly to a reservoir.